信号与系统

School of Electronic
and Information Engineering

Zhou Hang

2026.3

1. Introduction to Signal and System Analysis

- Introduction
- Signal description and classification
- System description and classification
- Applications of signals and systems

Description of the system

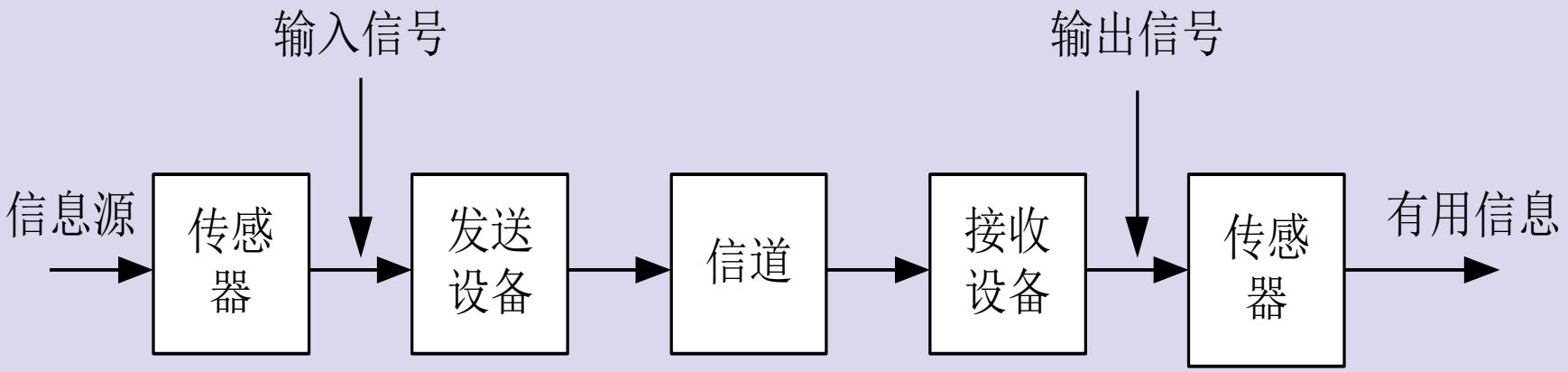
- ※ system definition
- ※ system in life
- ※ system description

1. system definition

- The word system in English comes from ancient Greek (systεmα), It means a whole made up of many parts.
- Bertalanffy, the founder of system theory, defined: "A system is a complex of interrelated and interacting elements".
- **This course defines**: A system is an organic whole composed of interacting and dependent units with specific functions.

1. system definition

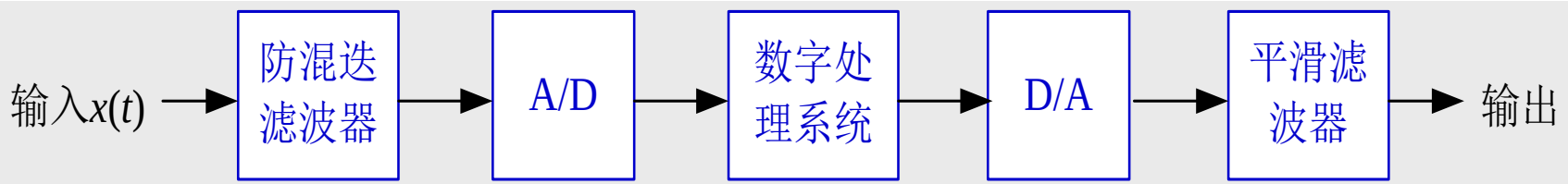
- A system is a whole that is composed of several units that interact and depend on each other and have specific functions.



电视广播通信系统框图

1. system definition

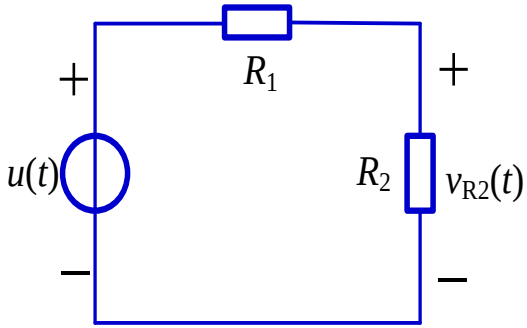
- A system is a whole that is composed of several units that interact and depend on each other and have specific functions.



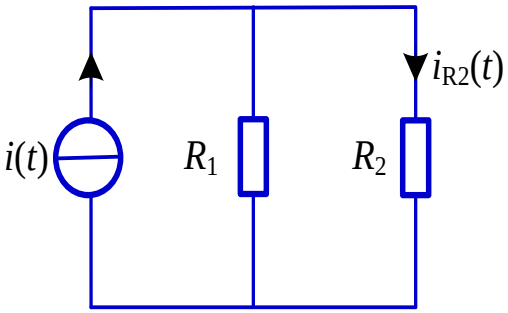
信号处理系统

2. Systems in life

- simple circuit system



voltage divider



bypass/shunt

$$v_{R2}(t) = \frac{R_2}{R_1 + R_2} u(t)$$

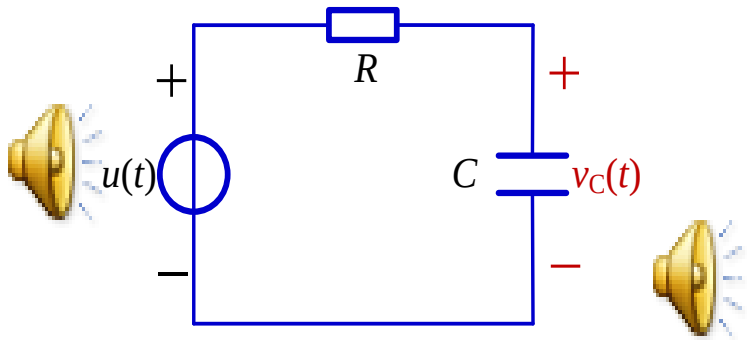
$$i_{R2}(t) = \frac{R_1}{R_1 + R_2} i(t)$$

2. Systems in life

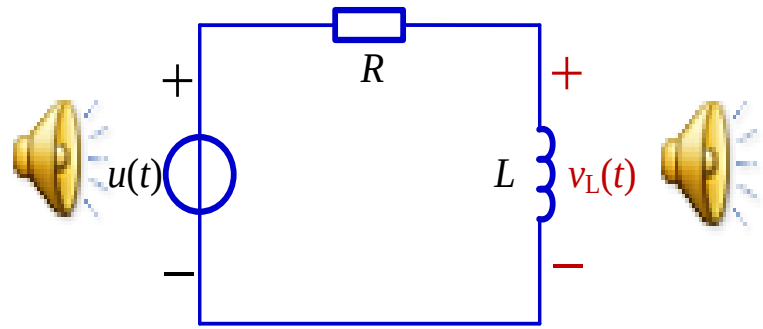
- simple circuit system

1=G $\frac{4}{4}$

$$1\ 2\ 3\ 4\ |\ 5\ 6\ 7\ \dot{1}\ |\ \dot{2}\ \dot{3}\ \dot{4}\ \dot{5}\ |\ \dot{6}\ \dot{7}\ \dot{1}$$



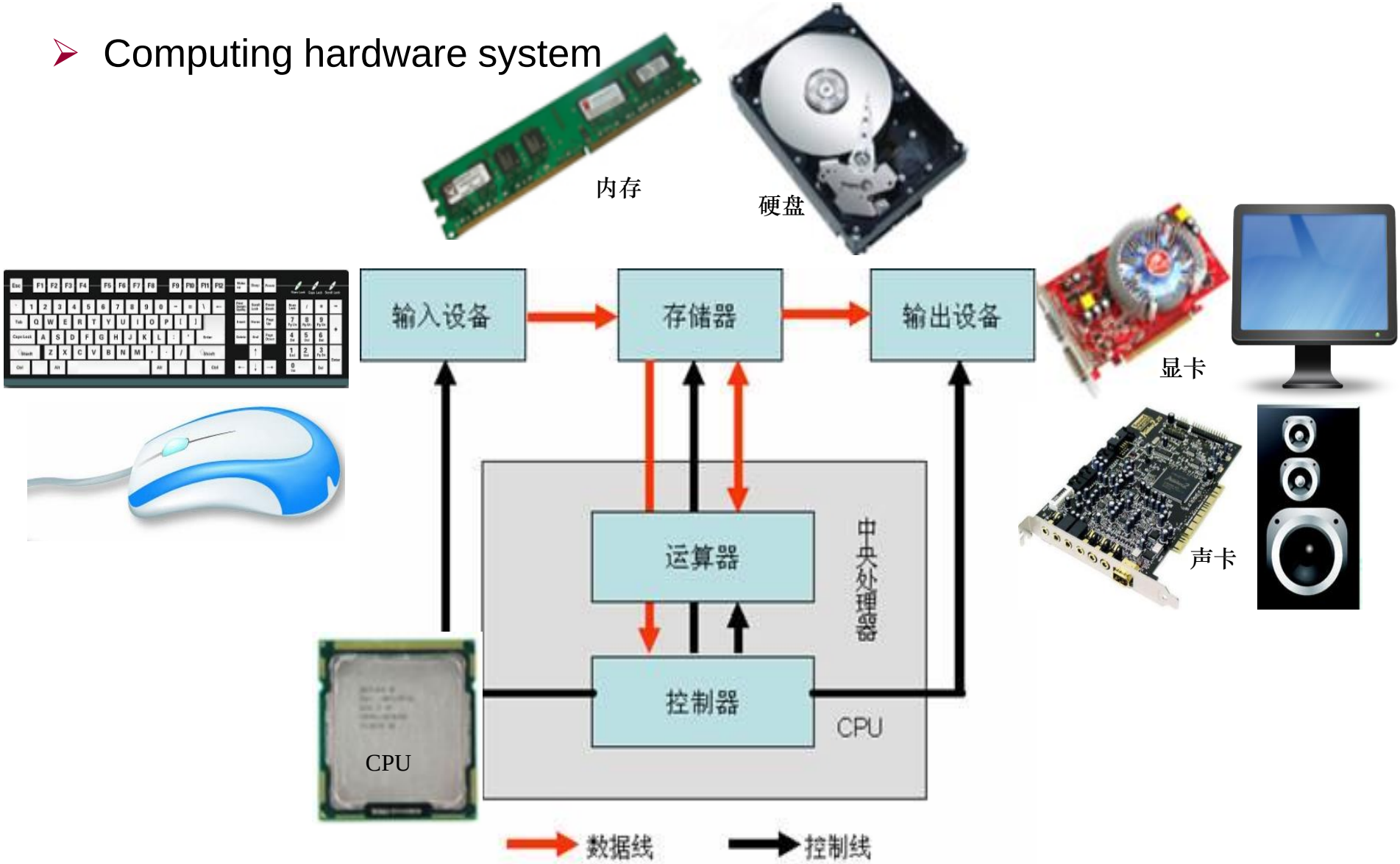
low pass filter
(Equalizer)



high pass filter
(Equalizer)

2. Systems in life

- Computing hardware system



2. Systems in life

- transportation system

  - Railway network

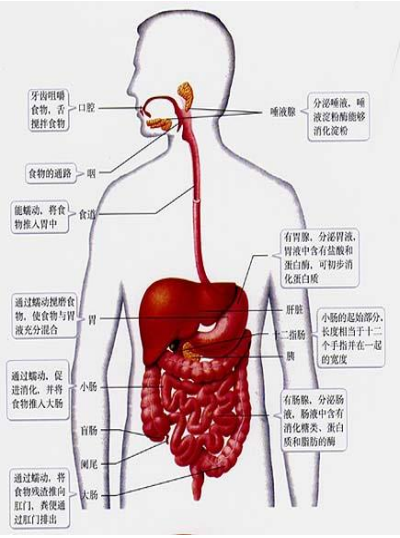
Eight vertical and eight horizontal:

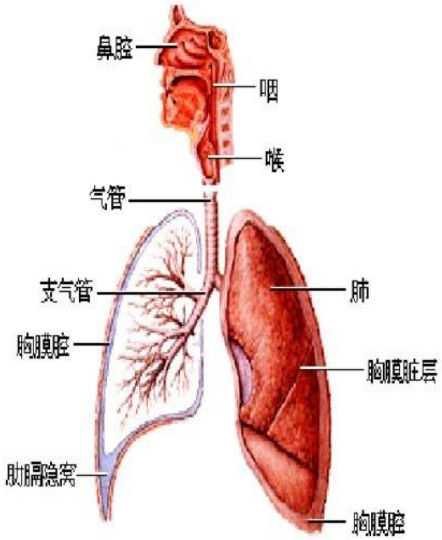
"Eight vertical" railway channel: Beijing-Harbin channel, coastal channel, Beijing-Shanghai channel, Beijing-Kowloon channel, Beijing-Guangzhou channel, Dazhan channel, Baoliu channel, and Lankun channel.

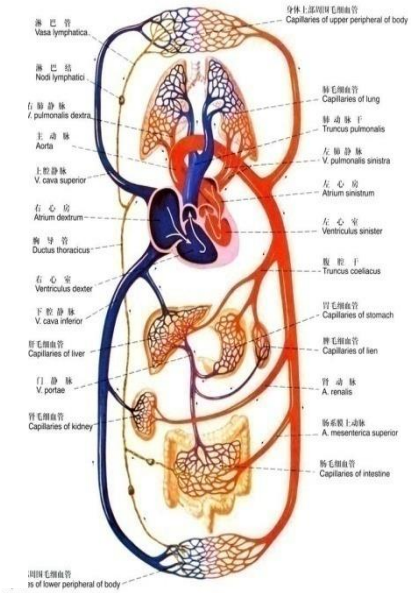
"Eight Horizontal" Railway Channel: Jinglan Passage, North Coal Transport Passage, South Coal Transport Passage, Taiyuan-Qingdao Passage, Land Bridge Passage, Ningxi Passage, Riverside Passage, Shanghai-Kunming (Chengdu) Passage, Southwest Sea Passage.

2. Systems in life

- biological systems - human body systems





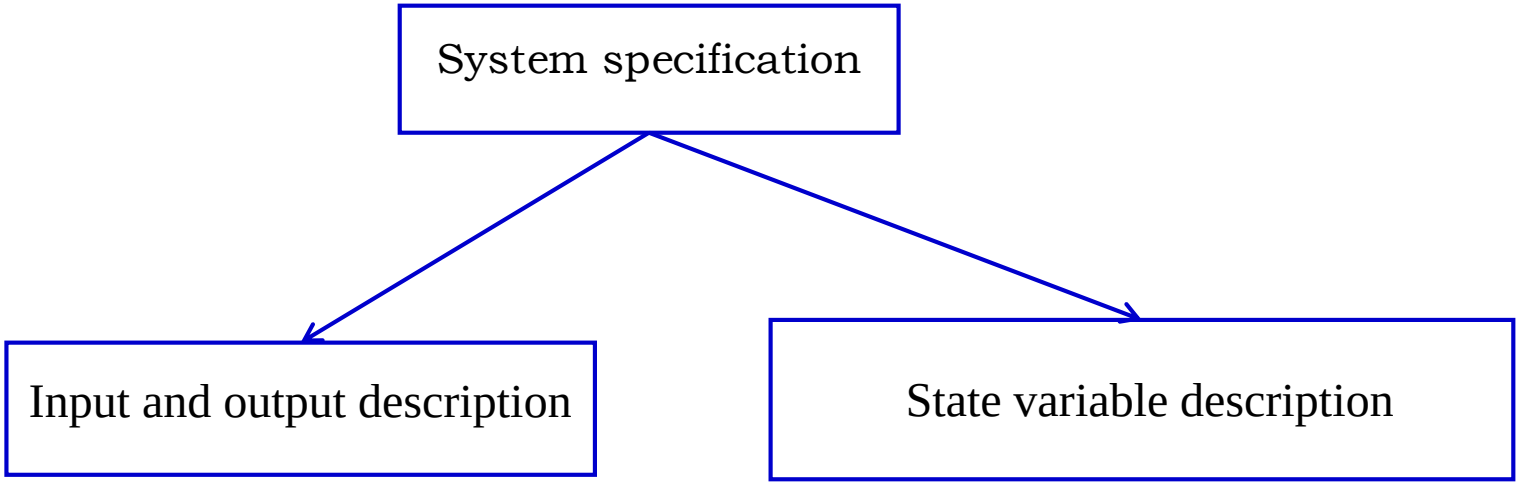


Digestive system Nervous system Respiratory system Circulatory system

Motor system, endocrine system, urinary system, reproductive system,…

3. system description

- **System description is to establish a mathematical model that describes the basic characteristics of the system**



Describe the system in terms of inputs and outputs external to the system

--External description

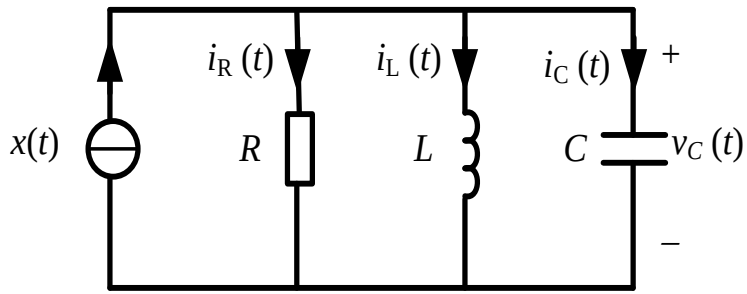
Through the internal state of the system

variable description system

--internal description

3. system description

➢Input and output description



RLC parallel circuit

If the input is x(t) and the output is v_C(t), then the relationship between input and output is

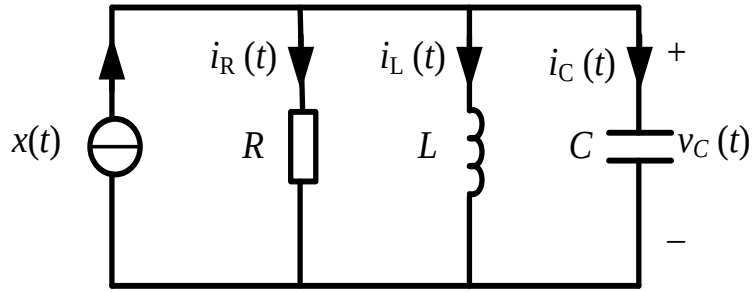
$$\boxed{\frac{\mathrm{d}v_C^2(t)}{\mathrm{d}t^2}+\frac{1}{RC}\frac{\mathrm{d}v_C(t)}{\mathrm{d}t}+\frac{1}{LC}v_C(t)=\frac{1}{C}\frac{\mathrm{d}x(t)}{\mathrm{d}t}}$$

Input and output description

Mathematical model of N-order system: N-order differential equation or N-order difference equation.

3. system description

➢State variable description



RLC parallel circuit

If the input is $x(t)$, calculating $i_R(t)$, $i_C(t)$, the relationship between the state variables $i_L(t)$, $v_C(t)$ and the input $x(t)$ can be established first

$$\frac{\mathrm{d}i_L(t)}{\mathrm{d}t} = \frac{1}{L} v_C(t)$$

$$\frac{\mathrm{d}v_C(t)}{\mathrm{d}t} = -\frac{1}{C} i_L(t) - \frac{1}{RC} v_C(t) + \frac{1}{C} x(t)$$

State variable description

Mathematical model of N-order system: N first-order differential equations or N first-order difference equations

Classification of the system

- ※ Continuous Time Systems vs. Discrete Time Systems
- ※ Linear and nonlinear systems
- ※ Time-invariant system and time-varying system
- ※ Causal and non-causal systems
- ※ stable system and unstable system

1. Continuous Time Systems vs. Discrete Time Systems

- Continuous-time systems:

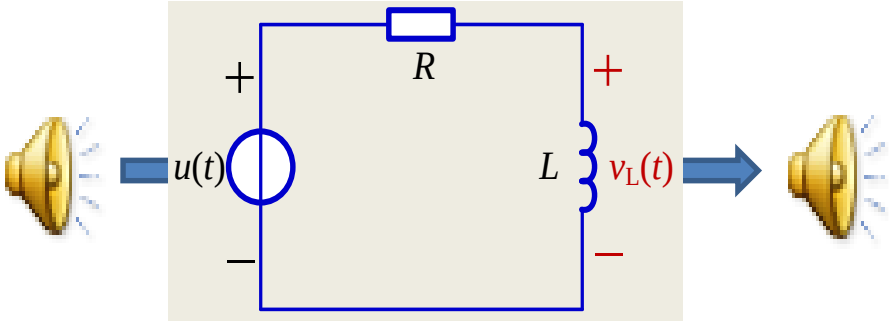
  The input stimulus and output response of the system are both continuous time signals.

- Discrete Time System:

  The input motivation and output response of the system are discrete time signals.

1. Continuous Time Systems vs. Discrete Time Systems

- Example of continuous time system and discrete time system



Continuous Time Systems

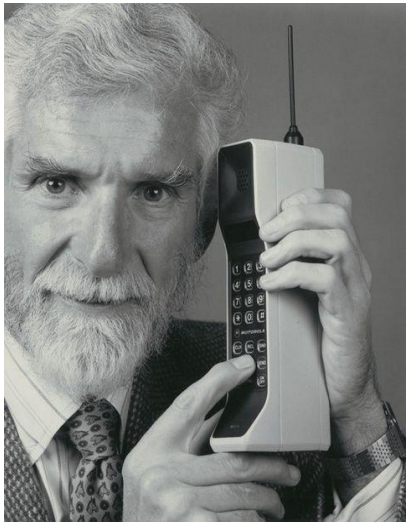
Discrete Time Systems

Circles consisting of simulated devices such as resistance, capacitance, inductance, diode, triode, simulation amplifier and other simulated devices

Circles consisting of digital components such as door circuits, triggers, counter, registers, encoders, decoders, comparators and other digital components

1. Continuous Time Systems vs. Discrete Time Systems

- Example of continuous time system and discrete time system



cell phone

smart phone

1G communication system (continuous time system)

3G communication system (discrete time system)

2. Linear system and non -linear system

- Linear system: systems with linear characteristics.

Uniform features and superposition features

Uniform features can be expressed as

$$\boxed{If\ x(t) \longrightarrow y(t),\ \ then\ Kx(t) \longrightarrow Ky(t)}$$

The superposition feature can be expressed as

If $x_1(t) \longrightarrow y_1(t),\ \ x_2(t) \longrightarrow y_2(t)$

The n $\boxed{x_1(t) + x_2(t) \longrightarrow y_1(t) + y_2(t)}$

Continuous System

2. Linear system and non -linear system

- Linear system: systems with linear characteristics.

Uniform features and superposition features

At the same time, it has uniform characteristics and superposition features to be linear characteristics, which can be expressed as

If $x_1(t) \longrightarrow y_1(t),\ x_2(t) \longrightarrow y_2(t)$

Then $\boxed{\alpha x_1(t) + \beta x_2(t) \longrightarrow \alpha y_1(t) + \beta y_2(t)}$

Among them, α 、 β are any constant

2. Linear system and non -linear system

- Linear system: systems with linear characteristics.

Uniform features and superposition features

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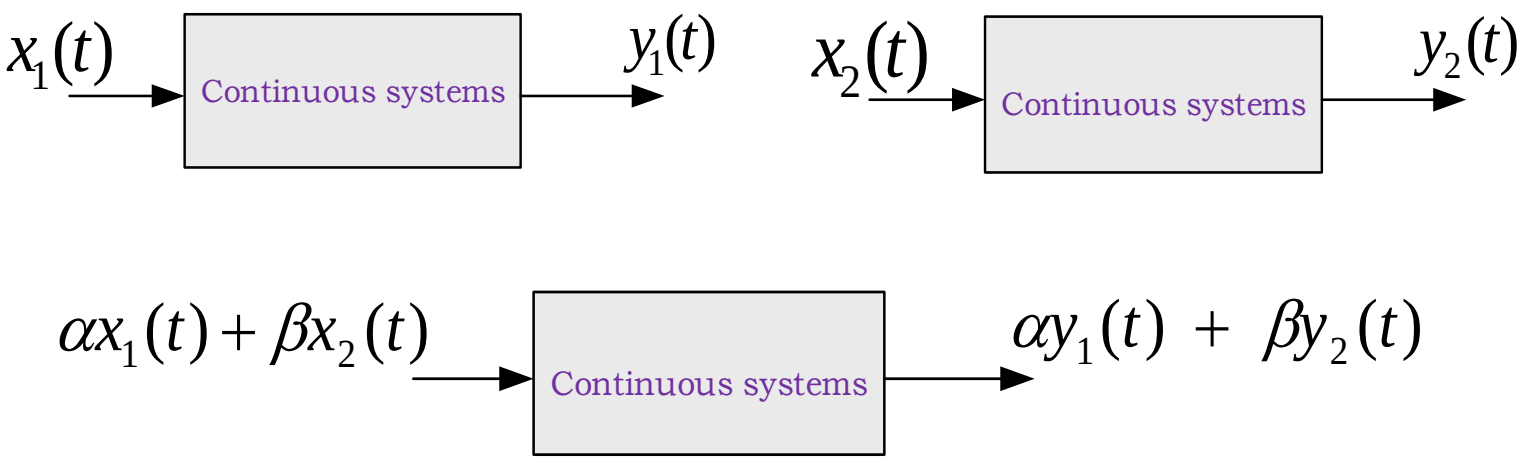

2. Linear system and non -linear system

- Linear system: systems with linear characteristics.

Uniform features and superposition features

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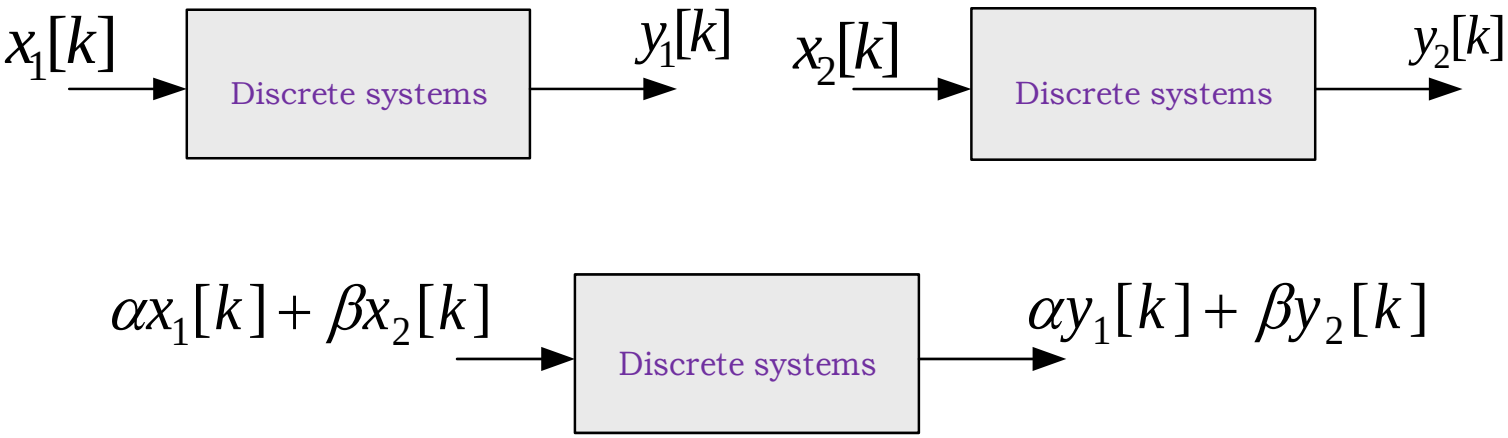


2. Linear system and non -linear system

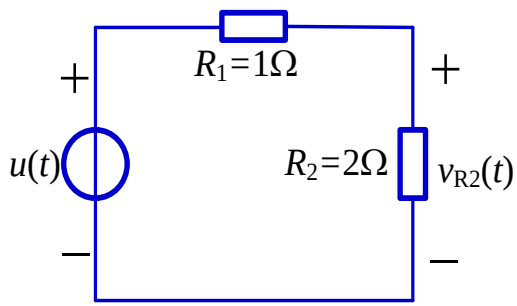
- Linear system: systems with linear characteristics.

Uniform features and superposition features

- Nonlinear System: A system that does not have linear properties.

2. Linear system and non -linear system

- Examples of linear systems



Voltage divider

$$v_{R2}(t)=\frac{R_2}{R_1+R_2}u(t)$$

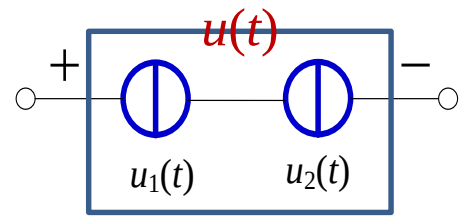
$$=\frac{2}{3}u(t)$$

When $u(t)=3\text{V}$ $\quad v_{R2}(t)=2\text{V}$

When $u(t)=2\times 3\text{V}$ $\quad v_{R2}(t)=2\times 2\text{V}$

satisfy Uniformity

When $u(t)=u_1(t)+u_2(t)$



When $u_1(t)$ acts alone, $v_{R2_1}(t)=\frac{2}{3}u_1(t)$

When $u_2(t)$ acts alone, $v_{R2_2}(t)=\frac{2}{3}u_2(t)$

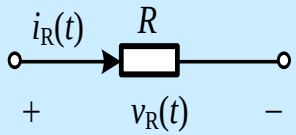
When the two act together,

satisfy superposition

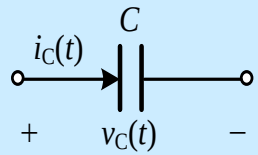
$$v_{R2}(t)=\frac{2}{3}u(t)=\frac{2}{3}[u_1(t)+u_1(t)]=v_{R2_1}(t)+v_{R2_2}(t)$$

2. Linear system and non -linear system

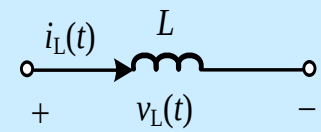
- Examples of linear systems



$$v_R(t) = Ri_R(t)$$



$$v_C(t) = \frac{1}{C}\int_{-\infty}^{t} i_C(\tau)\mathrm{d}\tau$$



$$v_L(t) = L\frac{\mathrm{d}i_L(t)}{\mathrm{d}t}$$

For resistors,let $i_R(t) = \alpha i_{R1}(t) + \beta i_{R2}(t)$

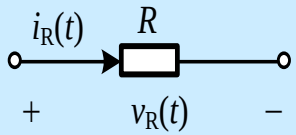
So $v_R(t) = R[\alpha i_{R1}(t) + \beta i_{R2}(t)] = \alpha R i_{R1}(t) + \beta R i_{R2}(t)$

$= \alpha v_{R1}(t) + \beta v_{R2}(t)$ have linear characteristics

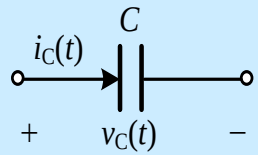
Resistors are linear elements

2. Linear system and non -linear system

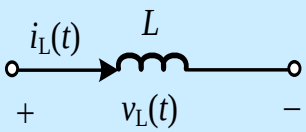
- Examples of linear systems



$$v_R(t) = Ri_R(t)$$



$$v_C(t) = \frac{1}{C}\int_{-\infty}^{t} i_C(\tau)\mathrm{d}\tau$$



$$v_L(t) = L\frac{\mathrm{d}i_L(t)}{\mathrm{d}t}$$

For capacitors, let $i_C(t) = \alpha i_{C1}(t) + \beta i_{C2}(t)$

$$\text{So } v_C(t) = \frac{1}{C}\int_{-\infty}^{t}[\alpha i_{C1}(\tau) + \beta i_{C2}(\tau)]\mathrm{d}\tau = \frac{\alpha}{C}\int_{-\infty}^{t} i_{C1}(\tau)\mathrm{d}\tau + \frac{\beta}{C}\int_{-\infty}^{t} i_{C2}(\tau)\mathrm{d}\tau$$

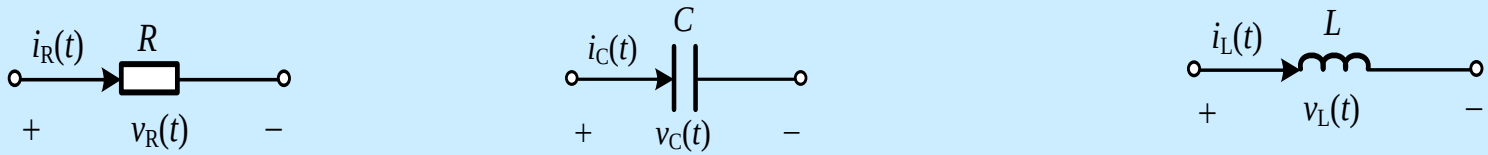
$$= \alpha v_{C1}(t) + \beta v_{C2}(t)$$

have linear characteristics

Capacitors are linear elements

2. Linear system and non -linear system

- Examples of linear systems



$$v_R(t) = Ri_R(t) \qquad v_C(t) = \frac{1}{C}\int_{-\infty}^{t} i_C(\tau)\mathrm{d}\tau \qquad v_L(t) = L\frac{\mathrm{d}i_L(t)}{\mathrm{d}t}$$

For inductors, let $i_L(t) = \alpha i_{L1}(t) + \beta i_{L2}(t)$

So $$v_L(t) = L\frac{\mathrm{d}[\alpha i_{L1}(t) + \beta i_{L2}(t)]}{\mathrm{d}t} = \alpha L\frac{\mathrm{d}i_{L1}(t)}{\mathrm{d}t} + \beta L\frac{\mathrm{d}i_{L2}(t)}{\mathrm{d}t}$$

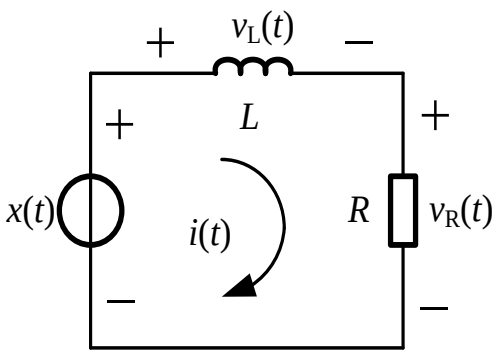
$$= \alpha v_{L1}(t) + \beta v_{L2}(t)$$ have linear characteristics

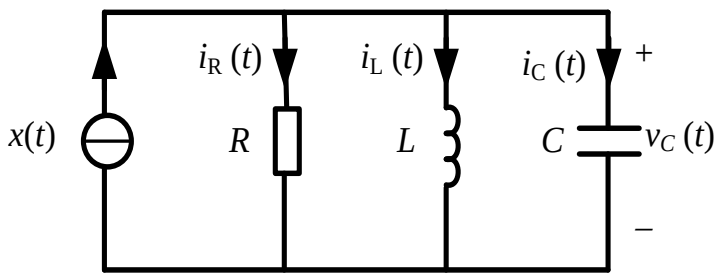
Inductors are linear components

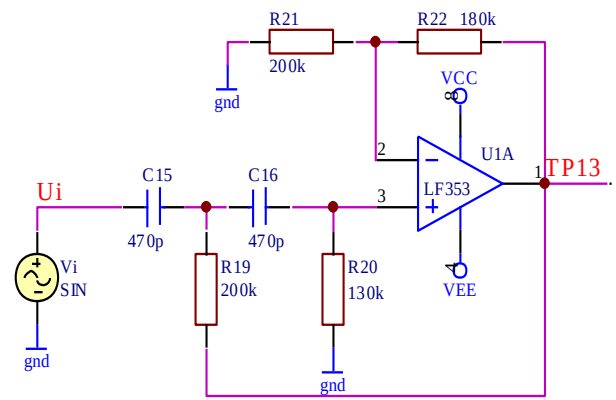
2. Linear system and non -linear system

➢ Examples of linear systems

A circuit consisting of linear elements (resistors, capacitors, inductors, etc.), independent sources, or linearly controlled sources is a linear system.







linear systems

2. Linear system and non -linear system

- [Example 1] It is known that the input *x(t)* and output *y(t)* of continuous time systems are bounded as follows.

$$(1)\ y(t)=3x(t)+4 \quad (2)\ y(t)=4x^2(t) \quad (3)\ y(t)=2\frac{\mathrm{d}x(t)}{\mathrm{d}t}+x(t)$$

Solution (1) Examining uniformity of properties

$$x(t)\rightarrow y(t)=3x(t)+4$$

$$Kx(t)\rightarrow 3Kx(t)+4\ \neq Ky(t)$$

does not satisfy the uniform characteristic, the system is nonlinear.

2. Linear system and non -linear system

- [Example 1]It is known that the input *x(t)* and output *y(t)* of the continuous time system are related as follows, try to determine whether these systems are linear systems.

$$(1)\ y(t)=3x(t)+4 \quad (2)\ y(t)=4x^2(t) \quad (3)\ y(t)=2\frac{\mathrm{d}x(t)}{\mathrm{d}t}+x(t)$$

Solution (2) Examining uniformity of properties

$$x(t)\rightarrow y(t)=4x^2(t)$$

$$Kx(t)\rightarrow 4[Kx(t)]^2=K^2[4x^2(t)]\neq Ky(t)$$

does not satisfy the uniform characteristic, the system is nonlinear.

2. Linear system and non -linear system

- [Example 1]It is known that the input *x(t)* and output *y(t)* of the continuous time system are related as follows, try to determine whether these systems are linear systems.

$$(1)\ y(t)=3x(t)+4 \quad (2)\ y(t)=4x^2(t) \quad (3)\ y(t)=2\frac{\mathrm{d}x(t)}{\mathrm{d}t}+x(t)$$

Solution (3) Examining uniformity of properties

$$x(t)\rightarrow y(t)=2\frac{\mathrm{d}x(t)}{\mathrm{d}t}+x(t)$$

$$Kx(t)\rightarrow 2\frac{\mathrm{d}Kx(t)}{\mathrm{d}t}+Kx(t)=K[2\frac{\mathrm{d}x(t)}{\mathrm{d}t}+x(t)]$$

$$=Ky(t)$$

Meet Uniformity characteristics

2. Linear system and non -linear system

- [Example 1]It is known that the input *x(t)* and output *y(t)* of the continuous time system are related as follows, try to determine whether these systems are linear systems.

$$(1)\ y(t)=3x(t)+4 \quad (2)\ y(t)=4x^2(t) \quad (3)\ y(t)=2\frac{\mathrm{d}x(t)}{\mathrm{d}t}+x(t)$$

Solution (3) Examine the superimposed characteristics

$$x_1(t)\to y_1(t)=2\frac{\mathrm{d}x_1(t)}{\mathrm{d}t}+x_1(t) \qquad x_2(t)\to y_2(t)=2\frac{\mathrm{d}x_2(t)}{\mathrm{d}t}+x_2(t)$$

$$x_1(t)+x_2(t)\to 2\frac{\mathrm{d}[x_1(t)+x_2(t)]}{\mathrm{d}t}+[x_1(t)+x_2(t)]$$

Meet Superposition characteristics

$$=[2\frac{\mathrm{d}x_1(t)}{\mathrm{d}t}+x_1(t)]+[2\frac{\mathrm{d}x_2(t)}{\mathrm{d}t}+x_2(t)]=y_1(t)+y_2(t)$$

Satisfying both the uniformity and superposition characteristics, the system is linear.

2. Linear system and non -linear system

- [Example 2] Determine whether integrator and delay are linear systems.

$$x(t) \longrightarrow \boxed{\int} \longrightarrow y(t)=\int_{-\infty}^{t} x(\tau)\mathrm{d}\tau \qquad x[k] \longrightarrow \boxed{\mathrm{D}} \longrightarrow y[k]=x[k-1]$$

Solution:

Assume $x(t)=\alpha x_1(t)+\beta x_2(t)$

then $y(t)=\int_{-\infty}^{t}[\alpha x_1(\tau)+\beta x_2(\tau)]\mathrm{d}\tau$

$$=\alpha\int_{-\infty}^{t} x_1(\tau)\mathrm{d}\tau+\beta\int_{-\infty}^{t} x_2(\tau)\mathrm{d}\tau$$

$$=\alpha y_1(t)+\beta y_2(t)$$

Therefore, it is a linear system

Assume $x[k]=\alpha x_1[k]+\beta x_2[k]$

then $y[k]=\alpha x_1[k-1]+\beta x_2[k-1]$

$$=\alpha_1 y_1[k]+\alpha_2 y_2[k]$$

Therefore, it is a linear system

Integrators and delayers are linear systems

3. Time-invariant system and time-varying system

- Time-invariant system: Under zero-state conditions, the relationship between the output response of the system and the input stimulus does not change with the starting point of the input stimulus acting on the system.

A time-invariant continuous-time system is expressed as

Zero State

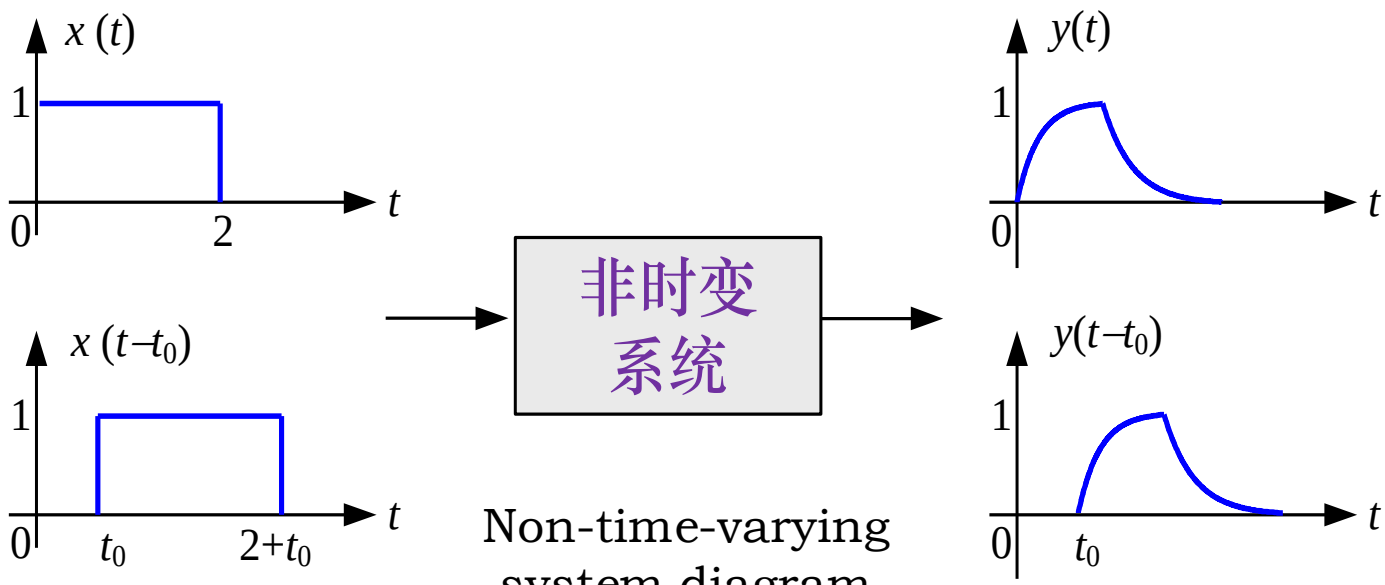
If $x(t) \longrightarrow y_{zs}(t)$ Then $x(t-t_0) \longrightarrow y_{zs}(t-t_0)$

A time-invariant discrete-time system is expressed as

If $x[k] \longrightarrow y_{zs}[k]$ Then $x[k-n] \longrightarrow y_{zs}[k-n]$

3. Time-invariant system and time-varying system

- Time-invariant system: Under zero-state conditions, the relationship between the output response of the system and the input stimulus does not change with the starting point of the input stimulus acting on the system.

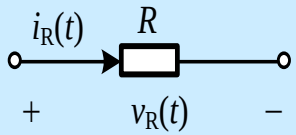


Non-time-varying system diagram

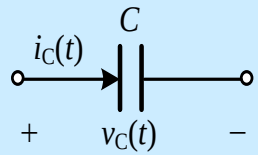
- Time-varying system: A system that does not satisfy the time-invariant property.

3. Time-invariant system and time-varying system

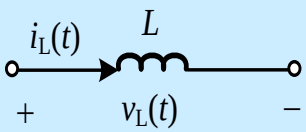
- Examples of time-invariant system



$$v_R(t) = Ri_R(t)$$



$$v_C(t) = \frac{1}{C}\int_{-\infty}^{t} i_C(\tau)\mathrm{d}\tau$$



$$v_L(t) = L\frac{\mathrm{d}i_L(t)}{\mathrm{d}t}$$

For resistors, if $x(t) = i_R(t - t_0)$

then $y(t) = Ri_R(t - t_0)$

$= v_R(t - t_0)$

time-invariant

If the resistance value is R(t), it changes with time, that is

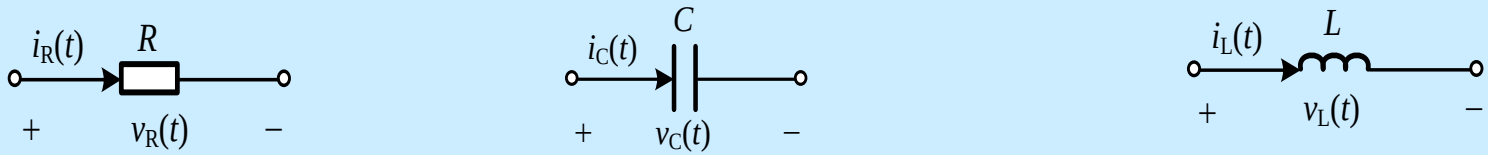
$v_R(t) = R(t)i_R(t)$, when the current is $x(t)$

$$y(t) = R(t)i_R(t - t_0) \neq v_R(t - t_0)$$

non time-invariant

3. Time-invariant system and time-varying system

- Examples of time-invariant system



$$v_R(t) = Ri_R(t) \qquad v_C(t) = \frac{1}{C}\int_{-\infty}^{t} i_C(\tau)\mathrm{d}\tau \qquad v_L(t) = L\frac{\mathrm{d}i_L(t)}{\mathrm{d}t}$$

For capacitors, if

$$x(t) = i_C(t - t_0)$$

$$\text{Then } y(t) = \frac{1}{C}\int_{-\infty}^{t} i_C(t - t_0)\mathrm{d}\tau$$

$$= \frac{1}{C}\int_{-\infty}^{t-t_0} i_C(t)\mathrm{d}\tau = v_C(t - t_0)$$

time-invariant

If the capacitance value is C(t), it changes with time, that is

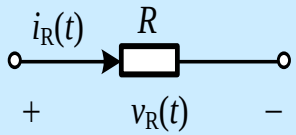
$$v_C(t) = \frac{1}{C(t)}\int_{-\infty}^{t} i_C(\tau)\mathrm{d}\tau$$

$$\text{Then } x(t) \rightarrow y(t) = \frac{1}{C(t)}\int_{-\infty}^{t-t_0} i_C(t)\mathrm{d}\tau$$

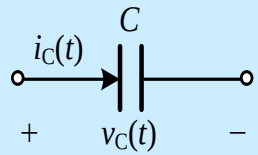
Time-varying $\neq v_C(t - t_0)$

3. Time-invariant system and time-varying system

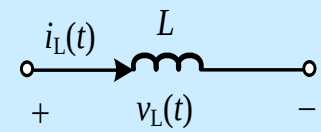
- Examples of time-invariant system



$$v_R(t) = Ri_R(t)$$



$$v_C(t) = \frac{1}{C}\int_{-\infty}^{t} i_C(\tau)\mathrm{d}\tau$$



$$v_L(t) = L\frac{\mathrm{d}i_L(t)}{\mathrm{d}t}$$

For inductors, if $x(t) = i_L(t - t_0)$

$$\text{So } y(t) = L\frac{\mathrm{d}i_L(t - t_0)}{\mathrm{d}t}$$

$$= v_L(t - t_0)$$

time-invariant

If the inductance value is L(t), it changes with time, that is

$$v_L(t) = L(t)\frac{\mathrm{d}i_L(t)}{\mathrm{d}t}$$

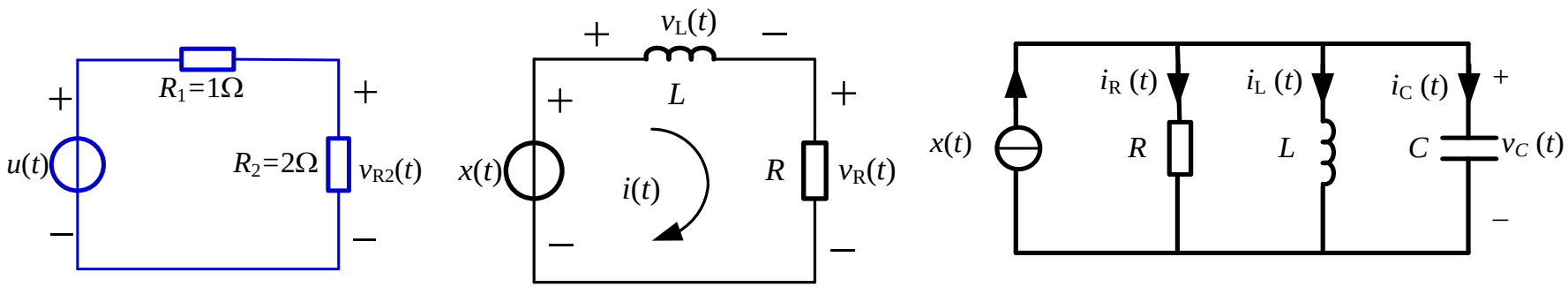
$$So,\ x(t) \rightarrow y(t) = L(t)\frac{\mathrm{d}i_L(t - t_0)}{\mathrm{d}t}$$

Time-varying $\neq v_L(t - t_0)$

3. Time-invariant system and time-varying system

- Examples of time-invariant system

A circuit whose component parameters do not change with time is a time-invariant system.



Time-invariant system

3. Time-invariant system and time-varying system

➢ [Example 3]

$x(t) \longrightarrow$ Continuous systems $\longrightarrow y(t)$ $\qquad$ $x[k] \longrightarrow$ Discrete systems $\longrightarrow y[k]$

The relationship between the input and output of the system is as follows. Try to

$$(1)\ y(t) = 2x(t) \qquad (2)\ y(t) = \cos t \cdot x(t) \qquad (3) y[k] = 2k \cdot x[k]$$

determine whether the system is time-invariant

Analysis: To determine whether a system is time-invariant, just need to find whether the system's zero-state response has the same delay when the input impulse is delayed, which is

$x(t-t_0) \longrightarrow$ Continuous systems $\longrightarrow y(t-t_0)$ $\qquad$ $x[k-k_0] \longrightarrow$ Discrete systems $\longrightarrow y[k-k_0]$

Note: When judging the time-invariant characteristics of the system, the initial state of the system is not involved.

3. Time-invariant system and time-varying system

- [Example 3]

$x(t) \longrightarrow$ Continuous systems $\longrightarrow y(t)$ $\qquad$ $x[k] \longrightarrow$ Discrete systems $\longrightarrow y[k]$

The relationship between the input and output of the system is as follows. Try to

$$(1)\ y(t) = 2x(t) \qquad (2)\ y(t) = \cos t \cdot x(t) \qquad (3)\ y[k] = 2k \cdot x[k]$$

determine whether the system is time-invariant

Solution:

(1) $x(t-t_0) \longrightarrow y_1(t) = 2x(t-t_0) = y(t-t_0)$ Time-invariant system

(2) $x(t-t_0) \longrightarrow y_1(t) = \cos t \cdot x(t-t_0) \neq y(t-t_0)$ Time-varying system

(3) $x[k-k_0] \longrightarrow y_1[k] = k \cdot x[k-k_0] \neq y[k-k_0]$ Time-varying system

Generally, if the input-output constraints of the system include variables related to t and k in addition to the input and output, it is a time-varying system.

3. Time-invariant system and time-varying system

- [Example 4] Try to determine whether a integrator, time delayer is a time-invariant system

$$x(t) \longrightarrow \boxed{\int} \longrightarrow y(t)=\int_{-\infty}^{t} x(\tau)\mathrm{d}\tau \qquad x[k] \longrightarrow \boxed{\mathrm{D}} \longrightarrow y[k]=x[k-1]$$

Solution:

Assume $x_1(t)=x(t-t_0)$

$$\Downarrow$$

thus $y_1(t)=\int_{-\infty}^{t} x(t-t_0)\mathrm{d}\tau$

$$=\int_{-\infty}^{t-t_0} x(t)\mathrm{d}\tau = y(t-t_0)$$

Assume $x_1[k]=x[k-k_0]$

$$\Downarrow$$

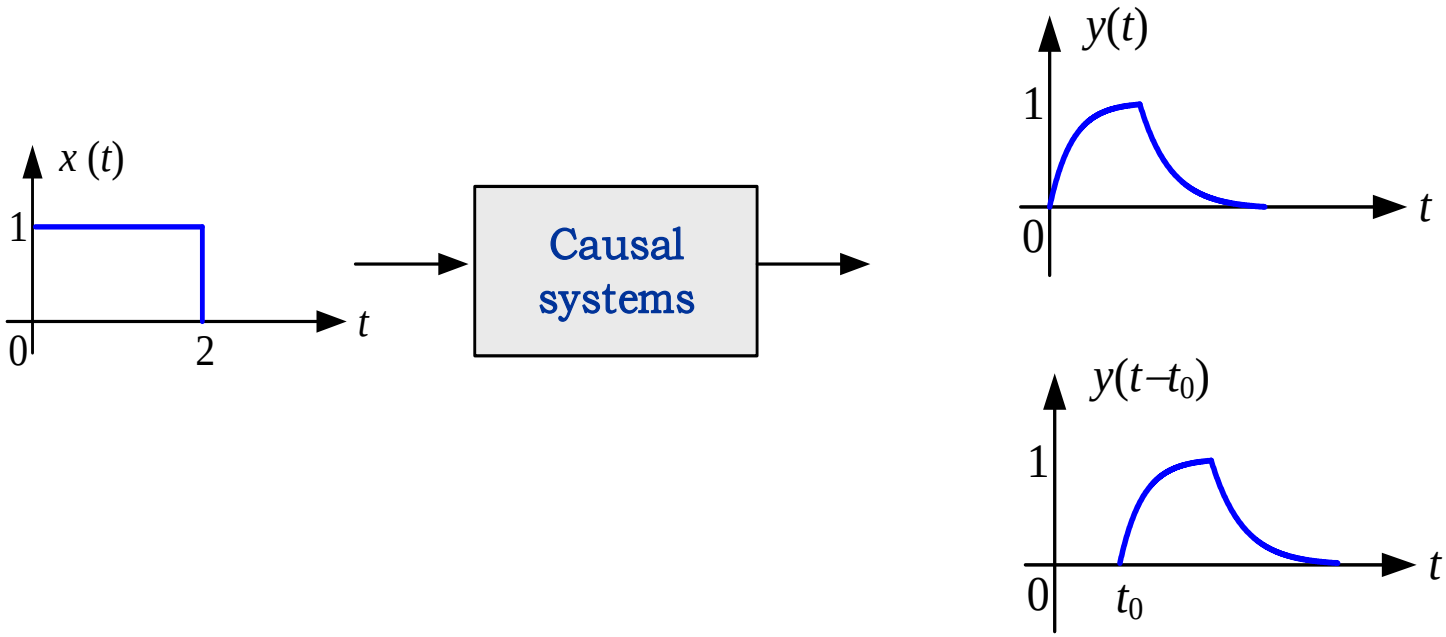
thus $y_1[k]=x[k-k_0-1]$

$$=y[k-k_0]$$

Integrators and delayers are time-invariant systems

4. Causal and non-causal systems

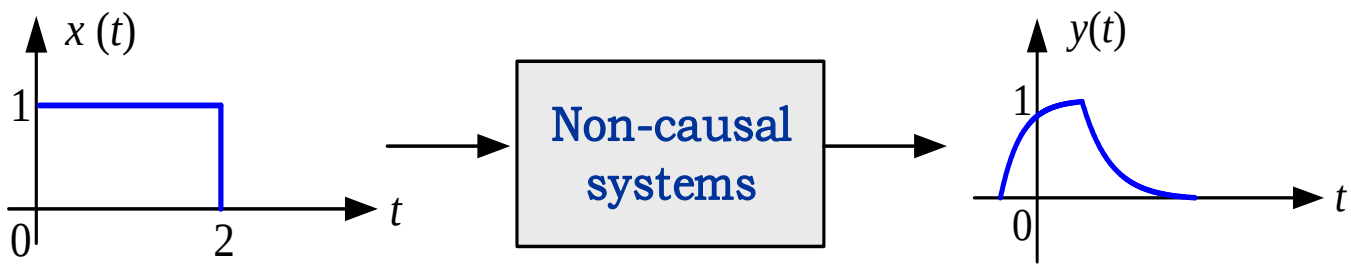
- Causal system:A system that produces an output response if and only if the input signal excites the system.



Schematic diagram of causal system

4. Causal and non-causal systems

- Non-causal system: A system that does not have causal properties is called non-causal system.



Schematic diagram of non-causal system

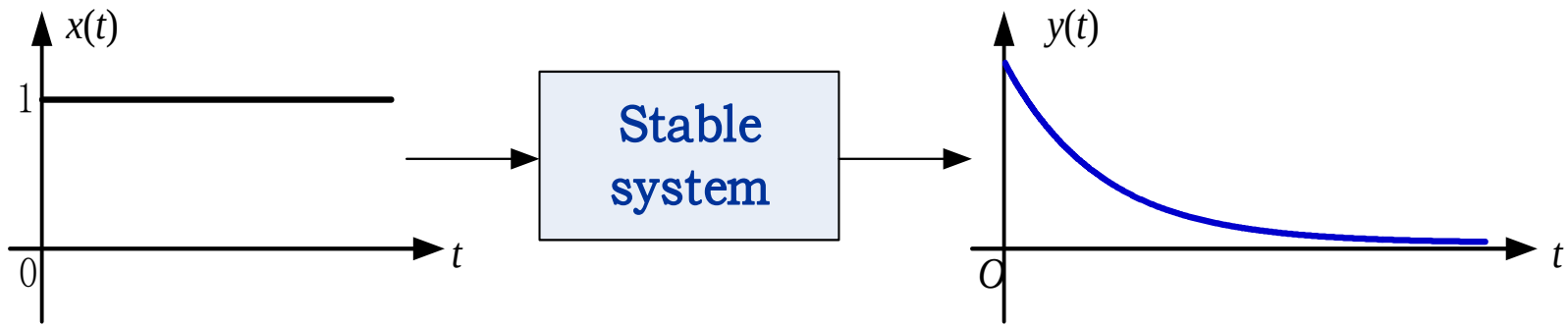
Physically realizable systems are causal systems

5. stable system and unstable system

➢ Stabilization System：

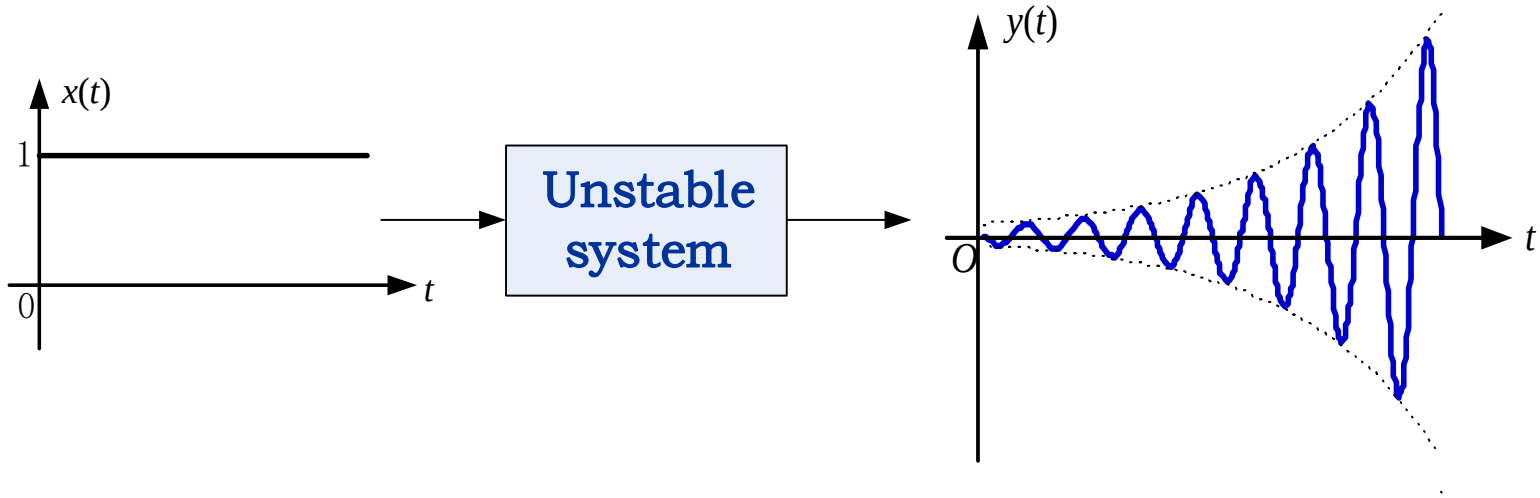
For any bounded input, the output of the system is also bounded. It is called BIBO stabilization system for short.

BIBO： Bounded Input, Bounded Output



5. stable system and unstable system

- Unstable system: A system with bounded inputs and unbounded outputs.



Thanks

Some of the materials quoted in this course are accumulated by the main lecturer over the years, from a variety of media and colleagues, peers and friends of the exchange, it is difficult to attribute them all, so I would like to state and thank you.